

4.1 Analytical Techniques

Question Paper

Course	CIEA Level Chemistry
Section	4. Analysis
Topic	4.1 Analytical Techniques
Difficulty	Medium

Time allowed: 40

Score: /31

Percentage: /100

Question 1a

Compound **X** contains atoms of carbon, hydrogen and oxygen only.

The mass spectrum of **X** is recorded. Information about the two peaks with m/e greater than 100 is shown in Fig. 1.1.

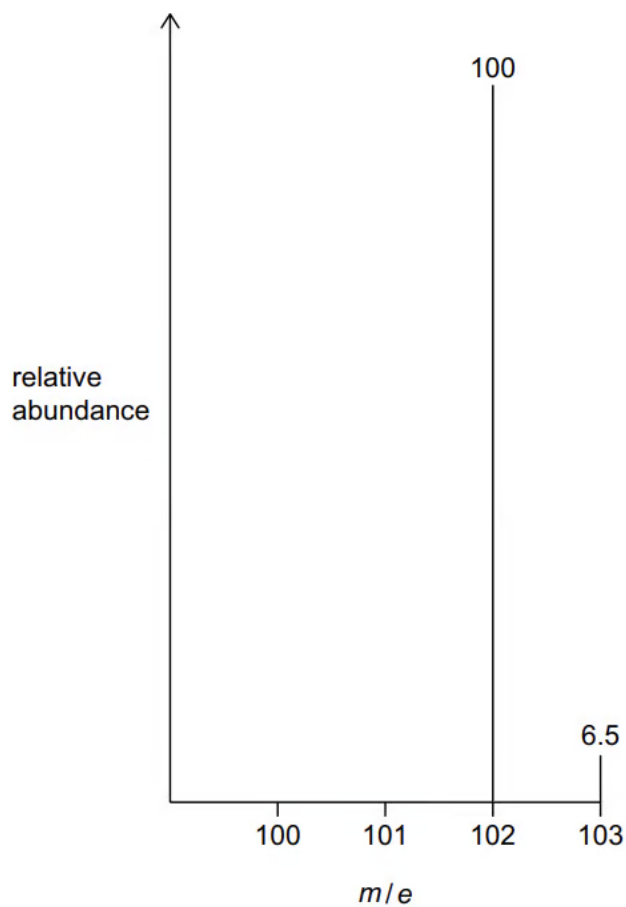


Fig. 1.1

A molecule of **X** contains 6 carbon atoms.

Demonstrate that this is correct using information from Fig 1.1. Show your working.

[2 marks]

Question 1b

Suggest the molecular formula of **X** using information from Fig. 1.1.

[1 mark]

Question 1c

Suggest the molecular formula of the fragment of **X** at $m/e = 31$.

[1 mark]

Question 1d

Fig 1.2 shows the infra-red spectrum of X.

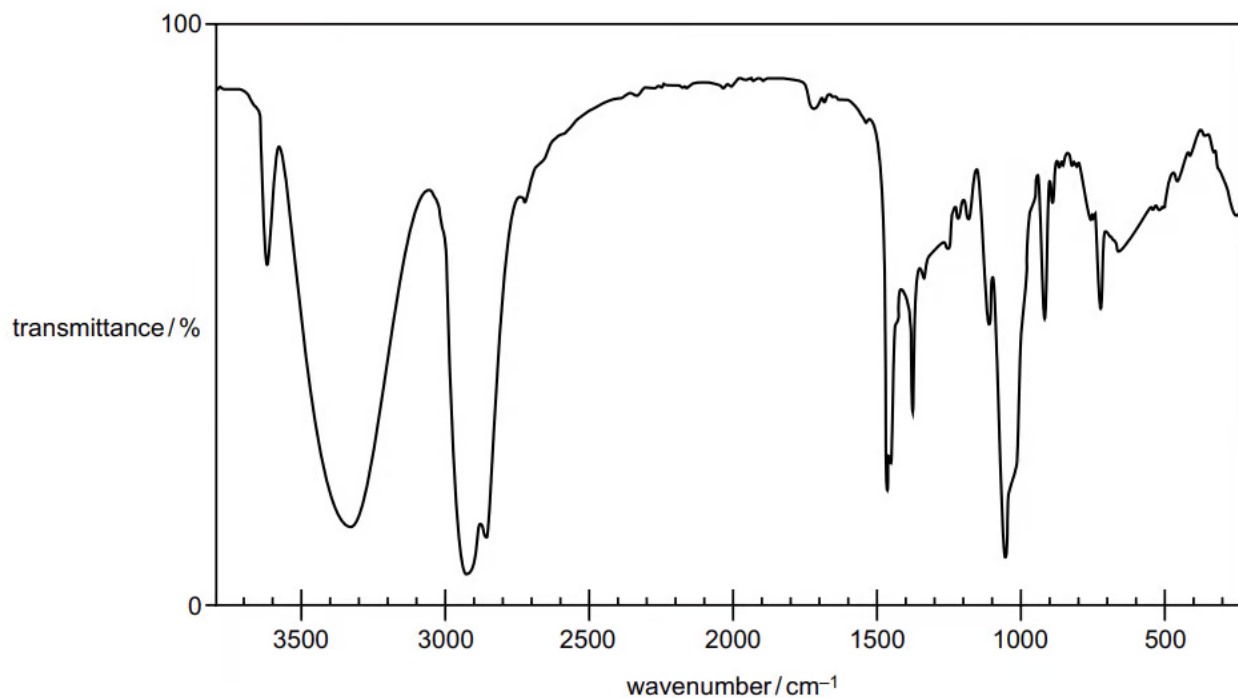


Fig. 1.2

The data in Fig 1.2, Table 1.1 and your answer to (b) should be used in answering this question.

Identify the functional group present in **X**. Explain your answer.

Table 1.1

Bond	Functional groups containing the bond	Characteristic infrared absorption range (in wavenumber) / cm^{-1}
C–O	hydroxy, ester	1040 – 1300
C=C	aromatic compound, alkene	1500 – 1680
C=O	amide	1640 – 1690
	carbonyl, carboxyl	1670 – 1740
	ester	1710 – 1750
C≡N	nitrile	2200 – 2250
C–H	alkane	2850 – 2950
N–H	amine, amide	3300 – 3500

O–H	carboxyl	2500 – 3000
	hydroxy	3200 – 3600

[1 mark]

Question 2a

The mass spectrum of magnesium is shown in Fig. 2.1.

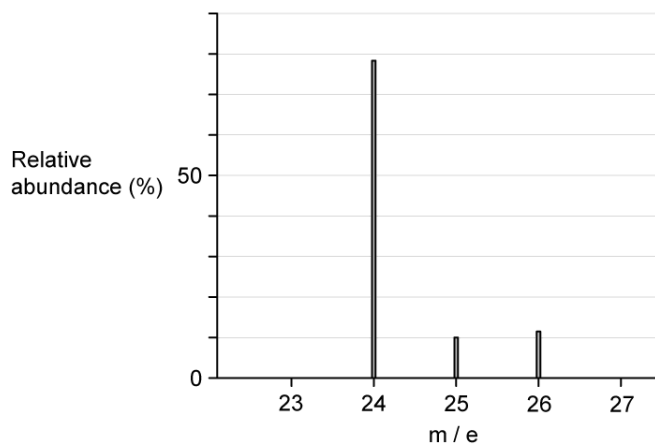


Fig. 2.1

From the mass spectrum, complete Table 2.1 with the relative abundances of the three isotopes.

Table 2.1

isotope	relative abundance
^{24}Mg	
^{25}Mg	
^{26}Mg	

[1 mark]

Question 2b

Use your values in (a) to calculate the relative atomic mass, A_r , of magnesium to two decimal places.

$$A_r(\text{Mg}) = \dots\dots\dots$$

[1 mark]

Question 2c

Give the full electronic configuration of the magnesium ion, $^{26}\text{Mg}^{2+}$.

[1 mark]

Question 2d

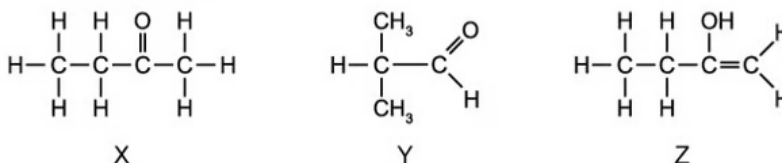
Boron has two naturally occurring isotopes, ^{10}B and ^{11}B . The relative atomic mass of boron is 10.8

Use this information to deduce the relative abundance of each isotope.

[3 marks]

Question 3a

A student wanted to distinguish between the following three compounds using mass spectrometry.



Explain why determining the exact mass using mass spectrometry would not help in distinguishing between the samples of X, Y and Z.

[2 marks]

Question 3b

Compounds **X**, **Y** and **Z** were analysed using IR spectroscopy. The spectrum of one of the compounds is shown in Fig 3.1.

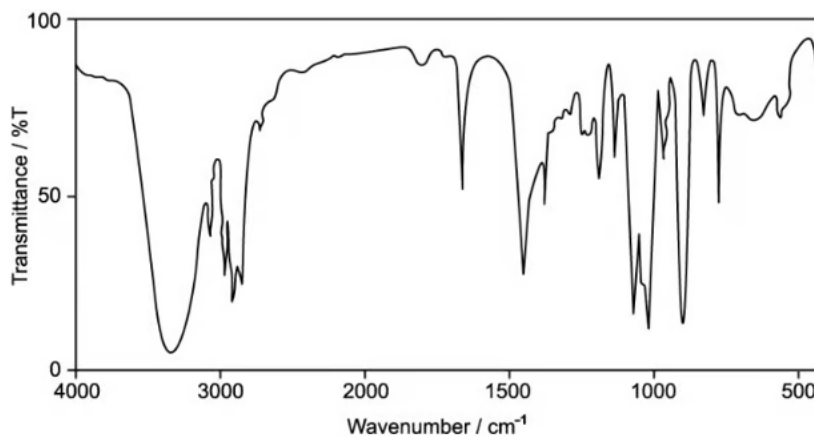


Fig 3.1

Identify which of the three compounds **X**, **Y** or **Z** this spectrum belongs to. Explain your reasoning.

[2 marks]

Question 3c

Explain why infrared spectroscopy alone could not be used to distinguish between compounds **X** and **Y**.

[2 marks]

Question 3d

Explain how mass spectrometry can distinguish between compounds **X** and **Y**.

[3 marks]

Question 4a

The isomers **A** and **B**, $C_5H_{10}O$, both form an orange precipitate when reacted with 2,4-DNPH.

A is unbranched and reacts with alkaline aqueous iodine to produce a yellow precipitate.

B does not react with alkaline aqueous iodine. It contains a chiral centre and produces a silver mirror when warmed with Tollens' reagent.

Draw the structure of the yellow precipitate produced by the reaction between **A** and alkaline aqueous iodine.

[1 mark]

Question 4b

i)

Give the structural formula of **A** and of **B**.

A

B

[2]

ii)

Explain the meaning of the term chiral centre.

[1]

[3 marks]

Question 4c

C and **D** are isomers with the molecular formula $C_3H_6O_2$. The infra-red spectra of isomers **C** and **D** are shown in Fig. 4.1 and Fig. 4.2.

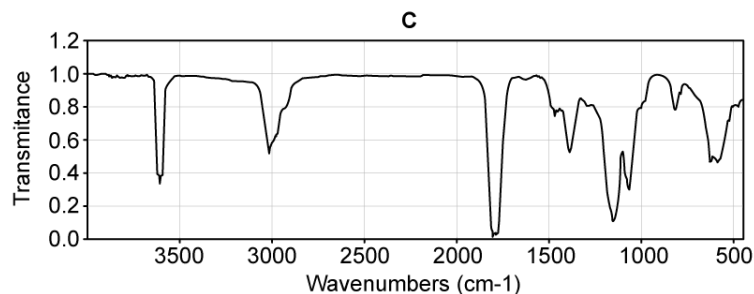


Fig. 4.1

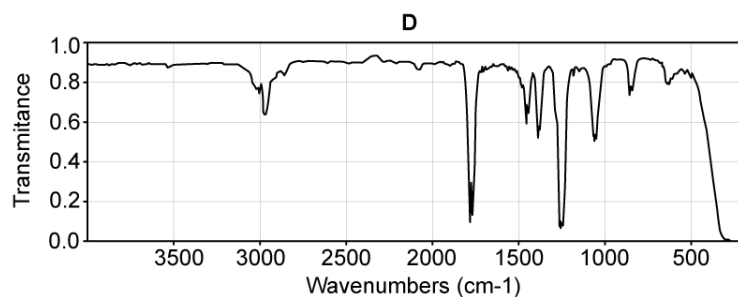


Fig. 4.2

Table 4.1

Bond	Functional groups containing the bond	Characteristic infrared absorption range (in wavenumbers)/cm ⁻¹
C–O	hydroxy, ester	1040 – 1300
C=C	aromatic compound, alkene	1500 – 1680
C=O	amide	1640 – 1690
	carbonyl, carboxyl	1670 – 1740
	ester	1710 – 1750
C≡N	nitrile	2200 – 2250
C–H	alkane	2850 – 2950
N–H	amine, amide	3300 – 3500
O–H	carboxyl	2500 – 3000
	hydroxy	3200 – 3600

Using Table 4.1, identify the bonds responsible for the principal peaks above 1500 cm^{-1} in each spectrum.

spectrum of **C**

spectrum of **D**

[2 marks]

Question 4d

Draw possible structures of isomers **C** and **D**

Isomer C	Isomer D

[2 marks]

Question 4e

Identify the ions responsible for peaks at 73, 45 and 29 in the mass spectrum for compound **C**.

[3 marks]

Model Answers: Medium

1a

a) It can be shown that compound **X** contains 6 carbon atoms by:

- $n = (100 / 1.1) \times (6.5 / 100)$; [1 mark]
- So $n = 5.91$ so there are 6 carbon atoms; [1 mark]

[Total: 2 marks]

- The peaks shown in the diagram are the 2 which are greater than 100, so the peak at $m/e = 102$ is the molecular ion peak and the peak at $m/e = 103$ is the $[M+1]$ peak caused by the carbon-13 isotope, which makes up for approximately 1.1% of all carbon atoms
- The number of carbon atoms, n , in a compound is deduced from the following formula:

$$n = \frac{100 \times \text{abundance of } [M+1]}{1.1 \times \text{abundance of } M^+ \text{ ion}}$$

- The abundance of $[M+1] = 6.5$ and the abundance of M^+ ion = 100, so:

$$n = \frac{100 \times 6.5}{1.1 \times 100} = 5.91$$

- so $n = 5.91$, rounding this gives 6 carbon atoms

1b

b) The molecular formula of **X** is:

- $102 - (6 \times 12) = 30$ so $C_6H_{14}O$; [1 mark]

[Total: 1 mark]

- The molecular ion peak occurs at a m/e value that corresponds to the molecular mass of the compound
- So, the molecular mass of compound **X** is 102

- It has 6 carbons which have a mass of $6 \times 12 = 72$
- The remaining mass must be due to hydrogen and oxygen
 - $102 - 72 = 30$
- Oxygen has a mass of 16, so there can only be 1 oxygen atom within the compound (as 2 oxygen atoms would have a combined mass of 32, which is too large)
- The remaining mass must be due to hydrogen atoms:
 - $30 - 16 = 14$
- As the mass of a hydrogen atom is 1, there must be 14 carbon atoms

1c

c) The molecular formula of the fragment of **X** at $m/e = 31$ is:

- $(\text{CH}_2\text{OH})^+$; [1 mark]

[Total: 1 mark]

- Think about some common fragments that could be found within a compound containing carbon, hydrogen and oxygen, and their m/e values:
 - $\text{CH}_3^+ = 15$
 - $\text{CH}_2^+ = 14$
 - $\text{OH}^+ = 17$
- Then think about what combination of these fragments would give an m/e value of 31
- CH_2OH^+ would give a value of $14 + 17 = 31$

1d

d) The functional group present in **X** is:

- A hydroxy group because of a (broad) peak at 3300
AND
One oxygen atom is present in the molecular formula; [1 mark]

[Total: 1 mark]

- The infra-red spectrum shows a broad peak at around 3300 cm^{-1} which corresponds to a hydroxyl group or carboxyl group
- The carboxyl group, $-\text{COOH}$, contains 2 oxygen atoms and the hydroxyl group, $-\text{OH}$,

contains 1 oxygen atom

- Compound X cannot contain the carboxyl group as it only has 1 oxygen atom, therefore it must contain a hydroxyl group

2a

a) The relative abundance of the isotopes is:

isotope	relative abundance
^{24}Mg	78-79
^{25}Mg	10
^{26}Mg	11-12

- All three values correct; [1 mark]

[Total: 1 mark]

- A range is given to allow for the readability of the diagram, but the values must add up to 100% to score the mark

2b

b) The relative atomic mass, A_r , of magnesium to two decimal places is:

- $(0.78 \times 24) + (0.10 \times 25) + (0.12 \times 26) = 24.34$; [1 mark]

[Total: 1 mark]

- To find the relative atomic mass, multiply the abundance by the isotopic mass for each isotope and sum the result
- Don't forget you need to convert the percentage to a decimal or the answer will need to be divided by 100

2c

c) The full electronic configuration of the magnesium ion, $^{26}\text{Mg}^{2+}$.

- $1s^2 2s^2 2p^6$; [1 mark]

[Total: 1 mark]

- Don't be fooled by the question asking you for the electronic configuration of an isotope – the isotopic mass is irrelevant
- You need to subtract two electrons from the atom to achieve the configuration of the +2 ion

2d

d) Deducing the relative abundance of each isotope of boron:

- $10.8 = \frac{10x + 11(100 - x)}{100}$

OR

$$10x + 1100 - 11x = 1080; \text{ [1 mark]}$$

- $\therefore x = 1100 - 1080 = 20; \text{ [1 mark]}$
- Isotope 1 / 10 relative abundance = 20

AND

$$\text{Isotope 2 / 11 relative abundance} = 80; \text{ [1 mark]}$$

[Total: 3 marks]

- If the relative atomic mass of boron is 10.8 we can take an average of the % abundances and make them equal to the relative atomic mass
- The two isotopes must have respective atomic masses of 10 and 11 as the relative atomic mass is 10.8
- We know that
- $A_r = \frac{(\text{relative abundance} \times \text{mass of isotope}) + (\text{relative abundance} \times \text{mass of isotope})}{100}$
- The relative abundances must add up to 100
- We can then put these numbers into the equation
- $10.8 = \frac{10x + 11(100 - x)}{100}$
- The isotope which has a mass of 10 has an abundance of 20%, as x is equal to 20
- The isotope which has a mass of 11 has an abundance of 80%, as $100 - x$ is equal to 80

3a

a) Using mass spectrometry to determine the exact M_r would not help distinguish between X, Y and Z because:

- All of the compounds have the same molecular formula

OR

All of the compounds are C_4H_8O

OR

All of the compounds have the same number and type of each atom; [1 mark]

- So they would all have the same / identical molecular mass / M_r shown on the mass spectrum

OR

The molecular ion peak / parent ion for all of the compounds would be the same; [1 mark]

[Total: 2 marks]

- The far right peak on the mass spectrum is the molecular ion peak
 - This is caused by the molecular ion (M^+) hitting the detector
- The molecular ion is the full molecule, but with a positive charge after it has been bombarded by high energy electrons and had an electron knocked off
- This peak tells you the M_r of the molecule

3b

b) The IR spectrum shows that the compound is:

- Compound **Z**; [1 mark]
- There is a peak at $3200 - 3600 \text{ (cm}^{-1}\text{)}$ from the OH bond / functional group

OR

There is a peak at $1620 - 1680 \text{ (cm}^{-1}\text{)}$ from the C=C bond / functional group; [1 mark]

[Total: 2 marks]

- You will always have the data sheet with the wavenumber ranges on, which you can use to analyse IR spectra
- Always use the exact ranges given on the data sheet in your answers

3c

c) Infrared spectroscopy could not be used to distinguish between compounds X and Y in part (b), because:

- They both have the same key functional group

OR

They both have a C=O bond and no other key functional group

OR

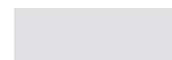
They are both carbonyls; [1 mark]

- Both spectra would have a peak at 1670 - 1740 (cm^{-1}) because of the C=O bond (and no other key distinguishing peak); [1 mark]

[Total: 2 marks]

- Infrared spectroscopy can be useful for identifying functional groups but like most analytical techniques, it is not definitive on its own

3d



3d

d) Mass spectrometry can distinguish between compounds **X** and **Y** because:

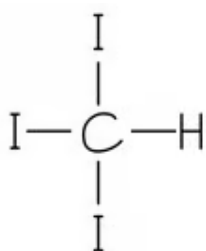
- They will give / produce different peaks in their mass spectra; [1 mark]
- Compound **X** can produce peaks at $m/e = 15, 29, 43$ and 57 ; [1 mark]
- Compound **Y** can produce peaks at $m/e = 13, 15, 28, 29, 42, 43$ and 57 ; [1 mark]

[Total: 3 marks]

- Fragments can be formed by breaking any of the major bonds within the compound structures

4a

a) The structure of the yellow precipitate is:



- Correct structure for tri-iodomethane; [1 mark]

[Total: 1 mark]

- The reaction between alkaline aqueous iodine to produce a yellow precipitate and compound **A** is known as the iodoform reaction
- The yellow precipitate indicates that compound **A** is a methyl ketone
- This precipitate is tri-iodomethane, CI_3H

4b

b)

i) The structural formula of **A** and of **B** are:

- **A** = $\text{CH}_3\text{CH}_2\text{CH}_2\text{COCH}_3$; [1 mark]

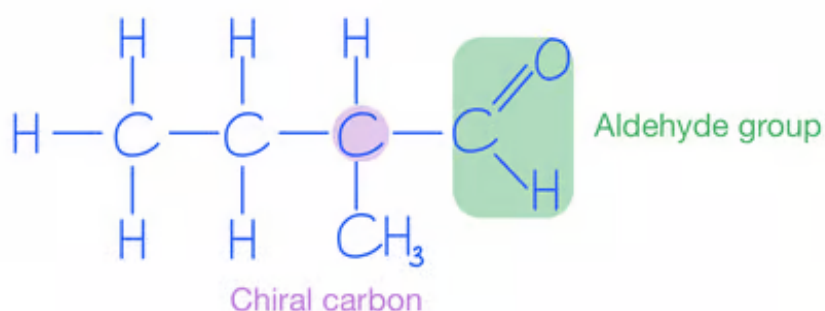
- **B** = $\text{C}_2\text{H}_5\text{CH}(\text{CH}_3)\text{CHO}$; [1 mark]

ii) A chiral carbon is:

- A (tetrahedral) atom with four different groups / atoms / substituents attached
OR
A carbon (atom) with four different groups / atoms / substituents attached; [1 mark]

[Total: 3 marks]

- You need to look at the clues in the question to help you identify compounds **A** and **B**
- They both react with 2,4-DNPH so will both contain a carbonyl group, -C=O
- In part (a) we have established that compound **A** has a straight chain and forms a yellow precipitate when reacted with alkaline aqueous iodine
 - The positive iodoform test means that the compound contains a methyl ketone group -COCH_3
 - So the carbonyl group must be on the second carbon atom
 - $\text{CH}_3\text{CH}_2\text{CH}_2\text{COCH}_3$
- Compound **B** contains a chiral carbon, so must be branched
- There is no reaction with iodoform so there can not be a methyl ketone group present
- The major clue to the identify of compound **B** is that it has a positive test with Tollens' reagent forming a silver mirror
 - This means that compound **B** is an aldehyde



4c

c)

i) The bonds responsible for the peaks are:

Compound C

- C=O (group / bond)

AND

O-H (group / bond); [1 mark]

Compound D

- C=O (group / bond)

AND

C-H (group / bond); [1 mark]

[Total: 2 marks]

- Make sure you identify the groups that correspond to the peaks above 1500 cm^{-1}

4d

d) Possible structures from **C** and **D** are:

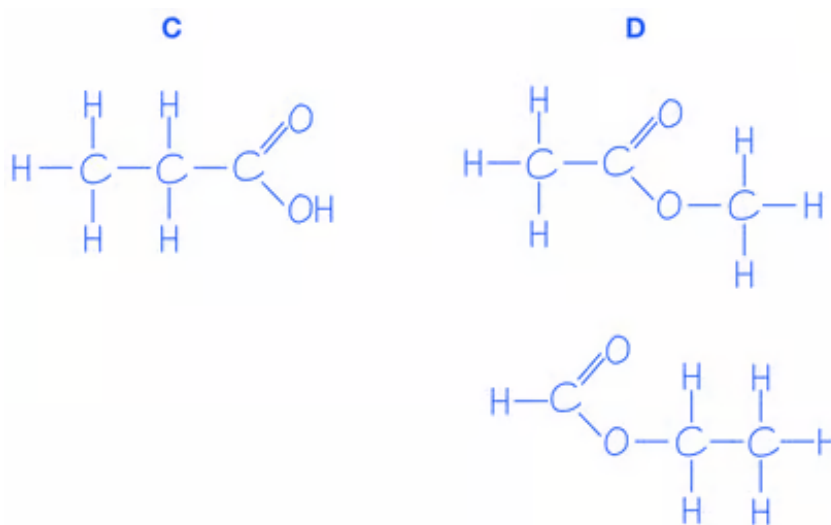
- **C**: Correct structure for propanoic acid; [1 mark]
- **D**: Correct structure for ethyl methanoate

OR

Methyl ethanoate; [1 mark]

[Total: 2 marks]

- Possible structures for **C** and **D** are



4e

e) The ions responsible for the peaks are:

- Peak at 73 = $[\text{C}_3\text{H}_5\text{O}_2]^+$; [1 mark]
- Peak at 45 = $[\text{COOH}]^+$; [1 mark]
- Peak at 29 = $[\text{C}_2\text{H}_5]^+$; [1 mark]

[Total: 3 marks]

- You are looking for the mass of the ion that corresponds to each peak
- The M_r of propanoic acid is 74
 - $\text{CH}_3\text{CH}_2\text{COOH}$
 - $\text{CH}_3 = 15$, $\text{CH}_2 = 14$, $\text{COOH} = 45$
 - $15 + 14 + 45 = 74$
- Therefore the peak at 73 must be $[\text{C}_3\text{H}_5\text{O}_2]^+$
- The peak at 45 must be $[\text{COOH}]^+$
- The peak at 29 must be $[\text{C}_2\text{H}_5]^+$ which includes the CH_3CH_2 group

